

Rule-Based AI Chatbot

Project 1 – Artificial Intelligence

Student Information

Name: Imran Ghafoor

Program: Bachelor of Science in Artificial Intelligence (BSAI)

Project Title: Rule-Based AI Chatbot

1. Introduction

Artificial Intelligence systems can be developed using different approaches. One of the most fundamental approaches is Rule-Based AI, where predefined rules are used to generate responses. This project focuses on building a simple chatbot that interacts with users using explicit rules and decision-making logic.

The chatbot accepts user input, processes it, and returns predefined responses based on a knowledge base dictionary. The project demonstrates the use of control flow, loops, dictionaries, and conditional statements in Python.

2. Project Objective

The objective of this project is to create a simple Rule-Based AI Chatbot that:

- Interacts with users continuously.
 - Responds to predefined inputs.
 - Uses a knowledge base dictionary.
 - Handles unknown inputs using a fallback response.
 - Terminates gracefully through exit commands.
-

3. Tools and Technologies

- Python 3.x
 - Visual Studio Code
 - Git & GitHub
-

4. Project Requirements

The chatbot was developed according to the following specifications:

1. Continuous input loop.
 2. Input sanitization.
 3. Knowledge base dictionary with 5 or more intents.
 4. Default fallback response.
 5. Exit strategy.
-

5. Methodology

Step 1: Knowledge Base Creation

A dictionary was created to store user intents and their corresponding responses.

Example:

```
knowledge_base = {  
    "hi": "Hello!",  
    "hello": "Hi there!",  
    "help": "How can I assist you?"  
}
```

Step 2: Input Sanitization

The user's input is cleaned before processing.

```
user_input = input("You: ").strip().lower()
```

This ensures consistent matching regardless of capitalization or extra spaces.

Step 3: Continuous Conversation Loop

The chatbot runs inside a while loop, allowing continuous interaction.

```
while True:
```

Step 4: Response Retrieval

Responses are retrieved using the dictionary `get()` method.

```
reply = knowledge_base.get(  
    user_input,  
    "Sorry, I don't understand that."  
)
```

The second argument acts as a fallback response when the entered command is not found.

Step 5: Exit Strategy

The chatbot terminates when the user enters predefined exit commands.

```
if user_input in ["bye", "exit", "quit"]:  
    break
```

6. System Workflow

1. Start chatbot.
2. Accept user input.
3. Sanitize input.
4. Check for exit command.
5. Search input in knowledge base.
6. Display response.
7. Repeat until user exits.

7. Features

- Rule-based decision making.
- Continuous user interaction.
- Input cleaning and normalization.
- Knowledge base dictionary.

- Default fallback mechanism.
 - Graceful exit handling.
 - Easy to expand with additional intents.
-

8. Sample Execution

User Interaction

```
You: hi
Bot: Hello! How can I help you?

You: help
Bot: I can answer simple predefined questions.

You: what is ai
Bot: Sorry, I don't understand that.

You: bye
Bot: Goodbye!
```

9. Learning Outcomes

After completing this project, the following concepts were learned:

- Control Flow
 - Loops
 - Dictionaries
 - Conditional Statements
 - Input Processing
 - Rule-Based Artificial Intelligence
 - Basic Chatbot Development
-

10. Conclusion

This project successfully demonstrates the implementation of a Rule-Based AI Chatbot using Python. The chatbot uses predefined rules stored in a knowledge base dictionary to generate responses. Input sanitization, fallback handling, and exit strategies ensure smooth interaction. The project provides a strong foundation for understanding how intelligent systems can be designed using explicit rules before progressing toward Machine Learning and Generative AI systems.